

(Lec - 32)

Array in JavaScript Assignment

PW EARNERS : Coding and Web Development

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Topics Covered

- Arrays
- push() and pop()
- shift() and unshift()
- splice() and slice()
- indexOf()
- find() and findIndex()
- flat()
- forEach()

Section 1 - Basic Array Operations

1. Create and Display an Array

Create an array containing the names of five fruits and display the complete array.

2. Add an Element Using push()

Create the following array and use push() to add "JavaScript" at the end.

Example:

Input: ["HTML", "CSS"]

Output: ["HTML", "CSS", "JavaScript"]

3. Remove an Element Using pop()

Remove the last element from the given array using pop().

Example:

Input: ["HTML", "CSS", "JavaScript"]

Output: ["HTML", "CSS"]

4. Remove the First Element Using shift()

Remove the first element from the following array using shift().

Example:

Input: ["Red", "Blue", "Green"]

Output: ["Blue", "Green"]

5. Add an Element at the Beginning

Use `unshift()` to add "HTML" at the beginning of the following array.

Example:

Input: ["CSS", "JavaScript"]

Output: ["HTML", "CSS", "JavaScript"]

6. Add Multiple Elements

Create an array containing two programming languages and use `push()` to add two more languages to the array.

Example:

Input: ["HTML", "CSS"]

After adding: "JavaScript" and "React"

Output: ["HTML", "CSS", "JavaScript", "React"]

Section 2 - `splice()` and `slice()`

7. Remove an Element Using `splice()`

Remove "CSS" from the following array using `splice()`.

Example:

Input: ["HTML", "CSS", "JavaScript", "React"]

Output: ["HTML", "JavaScript", "React"]

8. Add an Element Using `splice()`

Use `splice()` to add "CSS" between "HTML" and "JavaScript".

Example:

Input: ["HTML", "JavaScript"]

Output: ["HTML", "CSS", "JavaScript"]

9. Replace an Array Element

Use `splice()` to replace "Java" with "JavaScript".

Example:

Input: ["HTML", "CSS", "Java"]

Output: ["HTML", "CSS", "JavaScript"]

10. Extract Part of an Array

Use `slice()` to create a new array containing "CSS", "JavaScript", and "React".

Example:

Input: ["HTML", "CSS", "JavaScript", "React", "Node.js"]

Output: ["CSS", "JavaScript", "React"]

11. Create a Copy Using slice()

Create an array of your choice and use `slice()` without any arguments to create a copy of that array.

Section 3 - Searching in Arrays

12. Find the Index of an Element

Use `indexOf()` to find the index of "JavaScript".

Example:

Input: ["HTML", "CSS", "JavaScript", "React"]

Output: 2

13. Find an Element Using indexOf()

Create an array of programming languages and use `indexOf()` to find the position of "React".

14. Find an Object Using find()

Create an array of user objects containing name and age. Use `find()` to get the user whose name is "Rahul".

Example:

Input:

```
[  
  { name: "Rahul", age: 20 },  
  { name: "Priya", age: 22 }  
]
```

15. Find an Index Using findIndex()

Using an array of user objects, use findIndex() to find the index of the user whose name is "Priya".

Example:

```
[  
  { name: "Rahul", age: 20 },  
  { name: "Priya", age: 22 }  
]
```

Section 4 - Other Array Methods

16. Flatten a Nested Array

Use flat() to convert the nested array into a single-level array.

Example:

Input: [1, 2, [3, 4]]

Output: [1, 2, 3, 4]

17. Flatten a Multi-Level Array

Use flat() with an appropriate depth to flatten the following array completely.

Example:

Input: [1, [2, [3, 4]]]

Output: [1, 2, 3, 4]

18. Display Every Element Using forEach()

Create an array containing five colors and use forEach() to display every color.

19. Display Elements with Their Index

Create an array of programming languages and use forEach() to display each element along with its index.

Example Output:

0 HTML

1 CSS

2 JavaScript

20. Update an Array Using Multiple Methods

Perform the following operations on an array:

1. Add "React" using push().
2. Remove the first element using shift().
3. Display the final array.

Example:

Input: ["HTML", "CSS", "JavaScript"]

Final Output: ["CSS", "JavaScript", "React"]